SPETC v10: Physics of the Spectro-Photometric Exposure Time Calculator

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Abstract

SPETC is an exposure time calculator (ETC) for ground-based broad-band photometry and low- to high-resolution spectroscopy, descended from the original Fortran `interpola_spad` chain and rewritten in Python on `astropy`. This document derives every physical relation the code evaluates: the radiometric chain from a CALSPEC-calibrated template spectrum to detected photo-electrons; synthetic photometry in the SVO convention; atmospheric extinction, refraction, airmass and differential dispersion; the position-dependent night-sky model with van Rhijn airglow, zodiacal light, integrated starlight and Krisciunas–Schaefer moonlight; Gaussian and Moffat point-spread functions; the complete noise budget including scintillation and quantization; slit and slitless (Star Analyser) spectroscopy; and the treatment of extended sources. Every symbol used in the document is defined in a dedicated notation section. Version 10.2 adds the physics of the full audit of the code: full-bandpass slitless sky, resolution-invariant bin integration of templates, sky-subtraction noise, lunar distance and zodiacal-elongation dependence, airglow slant-path extinction, telluric bands, guiding blur, and radial-velocity/reddening template transforms. Expected performance and the known limits of validity are summarized.

Contents

1 Introduction

An exposure time calculator answers two reciprocal questions: what signal-to-noise ratio (S/N) a given exposure of a given target will reach, and how long an exposure is needed to reach a requested S/N. SPETC answers both for broad-band aperture photometry and for slit or slitless spectroscopy, for any telescope–detector combination the user describes, at any site, time and sky position.

The code descends from the Fortran `interpola_spad` programs (2007) and retains their template-calibration convention (Sect. 4.3), while the physics has been rebuilt in Python on `astropy` [Astropy Collaboration, 2022]: all radiometric quantities carry explicit units, all coordinate and time transformations

Table 1: Principal symbols.

Symbol	Meaning
F_λ	spectral flux density [erg s^{-1} cm^{-2} Å^{-1}]
A	unobstructed collecting area $\pi(D^2 - d^2)/4$
T_λ	filter transmission (0–1)
Q_λ	detector quantum efficiency (0–1)
η	scalar optics throughput excluding QE
O_λ	optional calibrated instrument response curve
$T_{\text{atm}}(\lambda)$	zenith atmospheric transmission
X	airmass
z	zenith distance; h apparent altitude
s	seeing FWHM [arcsec]
q	parallactic angle
S, B, D	source, sky, dark electrons in the aperture
σ_R	read noise per pixel [e$^-$ rms]
g	gain [e$^-$/ADU]
N_{pix}	pixels in the photometric aperture or extraction
R	resolving power $\lambda/\Delta\lambda$

use IERS-based ERFA routines, and every empirical relation is taken from the published literature cited in the section where it is used.

Version 10 upgrades the physical model in six areas, each described in its own section below:

1. a position-dependent night-sky model with van Rhijn airglow, zodiacal light, integrated starlight, and the Krisciunas and Schaefer [1991] moonlight model with the published scattering function (Sect. 6);

2. atmospheric refraction and the Pickering [2002] airmass (Sect. 5.4);

3. selectable Gaussian or Moffat [1969] point-spread functions (Sect. 7);

4. a noise budget extended with Young [1967] scintillation, Filippenko [1982] atmospheric dispersion and ADC quantization noise (Sects. 5.5, 8.1);

5. transmission-grating (Star Analyser 100/200) slitless spectroscopy (Sect. 9.3);

6. extended sources: galaxies, nebulae and planets treated as uniform surface brightness (Sect. 8.7).

Throughout, wavelengths are in Ångström internally, fluxes are F_λ in erg s^{-1} cm^{-2} Å^{-1}, and detected signals are in photo-electrons. The notation is summarized in Table 1.

2 The measurement chain: how the pieces fit together

Before the formula-by-formula derivations, this section tells the story once, end to end, so that each later section can be read as a close-up of one link in a chain you have already seen whole.

2.1 From a magnitude to a spectrum

The user gives the program one brightness number — "the target is magnitude 12.3 in V" — but a detector does not respond to magnitudes; it responds to photons per wavelength. The first task is therefore to turn one number into a full spectral energy distribution. This is what the *template* is for: an absolutely calibrated spectrum of a star of similar type supplies the *shape* $F_\lambda(\lambda)$, and the magnitude supplies the *level*. The connection between the two is the synthetic-photometry machinery of Sect. 4: the program computes what magnitude the template *would* have in the user's reference band, and rescales the whole spectrum by the ratio to the requested magnitude. Because the rescaling is total, the template's own flux units are immaterial — only its shape and its internal calibration matter — which is why CALSPEC standards, surface-brightness planet atlases and arbitrary-unit supernova models can all serve as templates (Sect. 4). The one number that anchors this step is the catalogue's m_{v0}, the magnitude the template file itself represents; it is the fulcrum on which the whole calibration turns.

2.2 From a spectrum above the atmosphere to electrons in a pixel

The calibrated spectrum then runs the gauntlet of the light path, each element a multiplicative wavelength-dependent factor inside one integral, Eq. (4):

$$F_\lambda \text{ (target, top of atmosphere)} \;\rightarrow\; T_{\text{atm}}(\lambda)^X \text{ (Sect. 5)} \;\rightarrow\; A\,\eta\,O_\lambda \text{ (telescope and optics)} \;\rightarrow\; T_\lambda \text{ (filter)}$$
$$\rightarrow\; Q_\lambda \text{ (detector QE)} \;\rightarrow\; \text{electrons s}^{-1}$$

The atmosphere enters twice more, in subtler roles than mere absorption: it *blurs* (the seeing PSF of Sect. 7, which decides how much light a finite aperture or slit captures and how bright the peak pixel gets), it *disperses* (differential refraction, Sect. 5.5, which drains blue light out of a misaligned slit), and it *twinkles* (scintillation, Sect. 5.7, a noise proportional to the signal itself). And the sky above the site is not dark: the sky model of Sect. 6 — airglow, zodiacal light, starlight, moonlight, twilight, or simply the user's SQM reading — adds its own electrons to every aperture and every resolution element, through the same instrument chain.

2.3 From electrons to a decision

The last link converts electron counts into the quantities an observer decides with. The CCD equation, Eq. (25), combines the source against every accumulated noise: its own shot noise, the sky's, dark current, read noise per pixel, scintillation, quantization. Solved forward it yields the S/N of a given exposure; inverted (Sect. 8.2) it yields the exposure for a requested S/N, and, when that exposure would saturate the brightest pixel — which the program predicts independently, from the unextracted PSF peak — the stack planner divides it into feasible frames. Spectroscopy adds one more translation: from S/N per resolution element to the equivalent-width uncertainty $\sigma(\text{EW})$ of Sect. 9.6, which states directly which spectral lines a night can measure. Everything downstream of the electron counts is exact bookkeeping; everything upstream is physics with stated approximations — the map of which is drawn in Sect. 11.

2.4 Where each ingredient comes from

Every factor in the chain is data the user controls, and each has a documented file format (User Guide, Sect. 6): the template from the catalogue; the filter transmission and zero point from an SVO profile (downloaded or synthesised from a measured curve, Sect. 10); the QE, optics response and gain table from the camera's datasheet; the atmospheric transmission from a site FITS table or the built-in broad-band fallback; the sky level from the physical model or a hand-held SQM meter. The design intent is that *nothing physical is hidden in the code*: what the program assumes, a file or a visible control states, and the run log in the Results window records.

3 Notation: every symbol, defined once

Every quantity used in this document is defined here, grouped by theme, with its units and the section where it first does physical work. Two deliberate collisions of astronomical convention are flagged explicitly: β denotes both the ecliptic latitude (sky model, Sect. 6) and the Moffat wing index (PSF, Sect. 7); α denotes both the lunar phase angle (Sect. 6) and the Moffat core radius (Sect. 7). The context — sky or PSF — always disambiguates them, and no equation mixes the two meanings.

Radiometry and magnitudes (Sect. 4)

Symbol	Units	Definition
λ	Å	Wavelength; all internal grids are in Å.
$F_\lambda(\lambda)$	$\mathrm{erg\,s^{-1}\,cm^{-2}\,Å^{-1}}$	Spectral flux density per unit wavelength (“F-lambda”), the quantity all templates are converted to.
F_ν	Jy	Spectral flux density per unit frequency; $1\,\mathrm{Jy} = 10^{-23}\,\mathrm{erg\,s^{-1}\,cm^{-2}\,Hz^{-1}}$.
Z_ν	Jy	The F_ν zero point of a magnitude system: the flux density of a zero-magnitude source (AB: 3631 Jy at every wavelength; Vega: per-filter values from the SVO profile).
m	mag	Magnitude: $m = -2.5\log_{10}(F/F_{\text{zero}})$ in the stated system and band.
m_{v0}	mag	Catalogue field: the visual magnitude represented by a stored template file as shipped (0 for BPGS spectra).
T_λ	—	Filter (passband) transmission, $0 \le T_\lambda \le 1$.
W_λ	—	Synthetic-photometry weighting: λ for a photon-counting detector, 1 for an energy integrator (SVO `DetectorType`).
Q_λ	—	Detector quantum efficiency: detected electrons per incident photon at λ.
O_λ	—	Optics/instrument throughput curve (everything between aperture and detector except the filter and QE).
η	—	Scalar (grey) efficiency factor for losses with no stated wavelength dependence.
A	$\mathrm{cm^2}$	Unobstructed collecting area, $A = \frac{\pi}{4}(D^2 - d_{\text{obs}}^2)$, with D the aperture and d_{obs} the central-obstruction diameter.
$h,\ c$	cgs	Planck constant and speed of light; hc/λ is the photon energy.
$\dot{S},\ \dot{B},\ \dot{d}$	$\mathrm{e^-/s}$	Detected electron rates: source (aperture- or element-integrated), sky, and dark current per pixel.
v	km/s	Radial velocity of the target; positive = receding.
$E(B-V)$	mag	Interstellar reddening (colour excess); $R_V = A(V)/E(B-V) = 3.1$ assumed; $A(\lambda)$ = extinction in magnitudes at λ (CCM89 law).

Geometry, time and the atmosphere (Sects. 5, 6)

Symbol	Units	Definition
h (altitude)	deg	Apparent altitude above the horizon (includes refraction unless stated "geometric").
z	deg	Zenith distance, $z = 90° - h$.
X	—	Airmass: the optical path through the atmosphere relative to the zenith path; computed from the apparent altitude with Eq. (5).
$T_{\mathrm{atm}}(\lambda)$	—	Zenith atmospheric transmission; the transmission at airmass X is T_{atm}^X (Bouguer law).
k_λ	mag/airmass	Extinction coefficient; $T_{\mathrm{atm}} = 10^{-0.4k_\lambda}$.
$n(\lambda)$	—	Refractive index of air at the site conditions.
$\Delta R(\lambda)$	arcsec	Differential atmospheric refraction (atmospheric dispersion) relative to a reference wavelength, Eq. (7).
$H,\ \varphi,\ \delta$	deg	Hour angle of the target, geographic latitude of the site, declination of the target.
q	deg	Parallactic angle: the position angle of the local vertical at the target, measured from North through East, Eq. (8).
$P,\ T_{\mathrm{site}},\ E$	hPa, °C, m	Site pressure, temperature (ISA defaults from the elevation) and elevation.
t_{exp}	s	Exposure time of a single frame.
d_{cm}	cm	Telescope aperture diameter (in the scintillation law).
$\sigma_{\mathrm{scint}}/I$	—	Fractional rms intensity modulation by scintillation, Eq. (10).
s, FWHM	arcsec	Seeing: the full width at half maximum of the seeing-limited stellar profile.
σ_g	arcsec	Guiding (tracking) error, rms per axis; adds a blur of FWHM $2.355\,\sigma_g$ in quadrature to the seeing.
r_0	m	Fried parameter of Kolmogorov turbulence; seeing $\propto \lambda/r_0$.

Sky brightness (Sect. 6)

Symbol	Units	Definition
μ	mag/arcsec^2	Surface brightness: the magnitude of the flux received from one square arcsecond of sky.
S10	—	Surface-brightness unit: the equivalent of one star of $m_V = 10$ per square degree; $\mu = 27.78 - 2.5\log_{10} q$ mag/arcsec^2 for a brightness of q S10 units.
nL	—	nanoLambert, the luminance unit of the Krisciunas–Schaefer moonlight model; converted to S10 by the factor 3.8.
$q_{\rm air}$, $q_{\rm zod}$, q_*	S10	Airglow, zodiacal-light and integrated-starlight brightness components.
a	—	Solar-cycle activity, interpolated between tabulated solar minima: 0.8 at minimum, 2.0 at maximum.
$V(z)$	—	van Rhijn airglow path-length factor, Eq. (12).
$\hat{h}$	km	Height of the airglow emitting layer (90 km).
$R_\oplus$	m	Mean Earth radius, 6.371×10^6 m.
β (sky)	deg	Ecliptic latitude of the field.
b	deg	Galactic latitude of the field.
ϵ	deg	Solar elongation of the field: its angular distance from the Sun.
α (Moon)	deg	Lunar phase angle: $0°$ = full Moon, $90°$ = quarter; computed as $180°$ minus the Sun–Moon elongation.
ρ	deg	Angular separation between the Moon and the target.
I^*	—	Lunar illuminance factor of the K&S model.
$f(\rho)$	—	K&S scattering function (Rayleigh + aureole), Eq. (15).
$X_{\rm m}$	—	Airmass of the Moon.
$d_{\rm moon}$	km	Topocentric distance of the Moon; the illuminance scales as $(\bar{d}/d_{\rm moon})^2$ with $\bar{d} = 384\,400$ km.
$\mu_{\rm SQM}$	mag/arcsec^2	Zenith V sky brightness measured by a Sky Quality Meter.

PSF and coupling (Sect. 7)

Symbol	Units	Definition
σ	arcsec	Gaussian width, $\sigma = \mathrm{FWHM}/2.355$.
α (Moffat)	arcsec	Moffat core radius, $\alpha = \mathrm{FWHM}/(2\sqrt{2^{1/\beta} - 1})$.
β (Moffat)	—	Moffat wing index; small β = strong wings; Gaussian is the $\beta \to \infty$ limit.
EE(r)	—	Encircled energy: fraction of the total PSF flux inside a circular aperture of radius r.
$T(w, \delta)$	—	Slit coupling: fraction of the PSF flux through an infinite slit of width w decentred by δ, Eq. (19).
F_ν (CDF)	—	Student-t cumulative distribution with ν degrees of freedom (only inside Eq. 19; not a flux).
p	arcsec/pixel	Plate scale: $p = 206265\, s_{\rm pix}/f_{\rm tel}$ with $s_{\rm pix}$ the pixel size and $f_{\rm tel}$ the telescope focal length in the same length units.

Noise budget and solvers (Sect. 8)

Symbol	Units	Definition
$S,\ B,\ D$	e^-	Accumulated source, sky and dark charge in the aperture over $t_{\rm exp}$.
$N_{\rm pix}$	—	Number of detector pixels summed in the aperture or extraction window.
σ_R	e^-	Read noise per pixel per frame, rms.
g	e^-/ADU	Conversion gain; the ADC quantization noise is $g/\sqrt{12}$ per pixel.
$N_{\rm bit}$	—	ADC bit depth; digital saturation at $g\,(2^{N_{\rm bit}}-1)$ electrons.
$n_{\rm sky}$	—	Number of pixels used to *measure* the background (sky annulus or slit sky windows); finite $n_{\rm sky}$ multiplies background variances by $(1+N_{\rm pix}/n_{\rm sky})$.
$\dot{b}$	e^-/s/pixel	Optional extra background rate (detector glow, stray light).
C	$s^{1/2}$	Scintillation constant of a configuration: $\sigma_{\rm scint}/I = C\,(2t_{\rm exp})^{-1/2}$.
Q	e^-/s	Total Poisson-like rate in the solver, $Q = \dot{S}+\dot{B}+\dot{D}$ (plus the scintillation rate).
Ω	arcsec^2	Angular area of an extended source.
$N,\ T$	—, s	Number of stacked frames and total integration time $T = N\,t_{\rm sub}$.

Spectroscopy (Sect. 9)

Symbol	Units	Definition
R	—	Resolving power, $R = \lambda/\Delta\lambda$.
$\Delta\lambda$	Å	Resolution element: the wavelength width the instrument cannot subdivide; one output row of the ETC.
$\mathcal{D}$	Å/pixel	Linear dispersion at the detector.
ℓ	pixel	Line-spread-function FWHM in dispersion-direction pixels (slitless: seeing $\oplus$ intrinsic $\oplus$ dispersion smear).
$n_{\rm mm}$	mm^{-1}	Grating groove density.
m (grating)	—	Diffraction order (1 for the Star Analyser).
L	mm	Grating-to-sensor distance of a converging-beam grating.
$f_{\rm coll},\ f_{\rm cam}$	mm	Collimator and camera focal lengths of a slit spectrograph.
$w_{\rm slit}$	arcsec	Slit width on the sky.
δx	Å	Width of one dispersion pixel, $\delta x = \Delta\lambda/(\text{pixels per element})$.
W	Å	FWHM of a spectral line in the Cayrel relation, Eq. (33).
EW	Å	Equivalent width: the width of a rectangle of continuum height with the same integrated absorption/emission as the line.

4 Radiometric framework

The radiometric layer solves one problem: converting between the languages of catalogues (magnitudes in named bands and systems) and the language of detectors (photo-electrons per second). The conversion must be exact in its conventions, because its errors are invisible — a wrong zero point or a magnitude-system confusion does not crash anything, it just biases every prediction by a constant factor. The layer therefore leans on three externally defined standards: the AB system's universal 3631-Jy zero point, the SVO Filter Profile Service's definition of a filter (transmission curve + zero point + detector-type flag, all in one file), and the CALSPEC absolute spectrophotometry that anchors the Vega scale. Everything else in the program consumes this layer's outputs and never touches a magnitude again.

4.1 Magnitude systems and zero points

A magnitude is a logarithmic flux ratio: $m = -2.5 \log_{10}(F/F_{\text{zero}})$, where F_{zero} is the flux of a zero-magnitude source *in the same system and band*. A magnitude system is therefore fully specified by its zero point. Systems are conventionally anchored in frequency space: a magnitude m in a system whose zero-magnitude source has the (frequency) flux density Z_ν (in jansky, $1\,\text{Jy} = 10^{-23}\,\text{erg}\,\text{s}^{-1}\,\text{cm}^{-2}\,\text{Hz}^{-1}$) corresponds to the wavelength flux density

$$F_\lambda(\lambda) \;=\; Z_\nu \, \frac{c}{\lambda^2} \, 10^{-0.4\,m}, \tag{1}$$

where the factor c/λ^2 is the Jacobian $|d\nu/d\lambda|$ that converts per-hertz to per-ångström flux, which the code evaluates with `astropy` units (`magnitude_f_lambda`). The AB system [Oke and Gunn, 1983] uses $Z_\nu = 3631\,\text{Jy}$ at every wavelength; Vega systems use the per-filter zero points delivered inside each SVO filter profile [Rodrigo and Solano, 2020, Bessell et al., 1998].

4.2 Synthetic photometry (SVO convention)

A *synthetic* magnitude is the magnitude a spectrum *would* have if measured through a filter — computed by integrating the spectrum against the filter response instead of observing it. For a spectrum F_λ and a filter of transmission T_λ (the fraction of light the filter passes at λ, between 0 and 1) it is

$$m \;=\; -2.5\,\log_{10} \frac{\displaystyle\int F_\lambda\, T_\lambda\, W_\lambda\, d\lambda}{\displaystyle\int F^0_\lambda\, T_\lambda\, W_\lambda\, d\lambda}, \qquad W_\lambda = \begin{cases} \lambda & \text{photon counter} \\ 1 & \text{energy integrator,} \end{cases} \tag{2}$$

where F^0_λ is the zero-magnitude spectrum of Eq. (1) and the weighting W_λ follows the SVO `DetectorType` flag of the profile: a photon-counting detector (CCD, CMOS) responds to the photon *number*, which for a given energy flux grows as λ (red photons are cheaper), so its integrand carries an extra factor λ; a bolometric/energy-integrating detector does not. Both conventions are implemented and regression-tested; for a coloured source they differ measurably.

4.3 Template calibration

Template spectra are absolutely calibrated CALSPEC/BPGS spectrophotometry [Bohlin et al., 2014]; the catalogue field m_{v0} records the visual magnitude represented by each file. The code first restores the visual-zero distribution, $F_\lambda \to F_\lambda\, 10^{0.4 m_{v0}}$ (the original Fortran convention), then scales it to the user's reference measurement:

$$F^{\text{cal}}_\lambda \;=\; F_\lambda\, 10^{0.4 m_{v0}}\; 10^{-0.4\,\left[m_{\text{ref}} - (m^{\text{syn}}_{\text{ref}} - m^{\text{syn}}_{V})\right]}, \tag{3}$$

where the synthetic colour term $(m^{\text{syn}}_{\text{ref}} - m^{\text{syn}}_{V})$ is computed from the template itself with Eq. (2) whenever the reference filter differs from the visual response, and is exactly zero in the original V-only case, which then reduces to the Fortran scale factor $10^{-0.4(V_{\text{target}} - m_{v0})}$.

4.4 Template sources

Templates are two-column ASCII or CALSPEC/STScI-style FITS binary tables (`WAVELENGTH`/`FLUX`, TUNIT-aware), which admits the whole STScI reference-atlas family: the CALSPEC standards, the solar-system surface-brightness atlas, the galactic emission-line atlas, the transient templates and the CLOUDY planetary-nebula grids. On import, the synthetic Vega V magnitude of each file is computed against the shipped Bessell.V profile and stored as the catalogue m_{v0}; since the ETC rescales every template to the entered target magnitude, files with arbitrary or surface-brightness flux units remain fully usable — only the spectral shape and the self-consistent m_{v0} matter. Templates that do not fully cover the V response carry an approximate m_{v0} and should be used with V as the reference filter (the exact same-response path).

4.5 Radial velocity and interstellar reddening

A template can optionally be transformed before calibration to represent a moving or reddened star. A radial velocity v (positive receding) shifts every wavelength by $\lambda \to \lambda\,(1 + v/c)$ — for $v = 300$ km/s a line at 5500 Å moves by 5.5 Å, comparable to a low-resolution element, so the shift matters exactly where line diagnostics are attempted. Interstellar dust attenuates the flux by $10^{-0.4\,A(\lambda)}$ with the Cardelli et al. [1989] (CCM89) extinction law $A(\lambda) = E(B-V)\,R_V\,[A(\lambda)/A(V)]$, where the colour excess $E(B-V)$ is the entered reddening, $R_V = 3.1$ is the standard diffuse-ISM ratio of total to selective extinction, and the bracketed shape function is CCM89's piecewise polynomial in $1/\lambda$. Both transforms are applied *before* the calibration of Sect. 4.3, which is the physically correct order: the observed magnitude of a reddened star already contains its reddening, so the transformation must change the template's *colours* while the calibration re-imposes the observed *level*.

4.6 Detected electron rate

The end product of the layer is the rate of detected photo-electrons. Every term of the chain now has a defined meaning: A is the collecting area (Sect. 3), η the scalar efficiency, T_λ the observing-filter transmission, Q_λ the detector quantum efficiency, O_λ the optics throughput curve, T_{atm}^X the atmospheric transmission at airmass X (Sect. 5), and λ/hc converts energy flux to photon number. The rate is

$$\dot{S} \;=\; A\,\eta \int F_\lambda^{\text{cal}}\,T_\lambda\,Q_\lambda\,O_\lambda\,T_{\text{atm}}(\lambda)^X\,\frac{\lambda}{hc}\,d\lambda. \tag{4}$$

The quadrature is trapezoidal on the *union* of the filter grid, the template grid and a dense baseline grid, which makes it exact for the piecewise-linear tabulated inputs: a narrow template emission line falling between two samples of a coarse filter table is integrated, not skipped (v10.2). In spectroscopy the same integral is evaluated per resolution element by cumulative integration (Sect. 9.1).

Coverage policy: on the *calibration* side (the reference band that sets the flux scale) full coverage is a hard requirement, because a truncated calibration integral silently rescales the whole spectrum. On the *observing* side (QE, observing filter, template beyond its range) missing coverage is zero-filled: truncation there only loses counts, and both engines apply the same rule.

5 Atmosphere

The atmosphere acts on a ground-based measurement in four physically distinct ways, and SPETC models each separately: it *absorbs and scatters* light out of the beam (extinction), it *bends* the beam (refraction, which changes the apparent altitude and, through its wavelength dependence, disperses the image into a tiny vertical spectrum), it *blurs* the image (seeing, treated in Sect. 7), and it *modulates* the total received flux on millisecond timescales (scintillation). Confusing these roles is the commonest source of error in home-made exposure calculators — e.g. applying an extinction correction twice, or ignoring scintillation because "the photon noise is already counted". This section fixes the conventions for each.

5.1 Extinction

Zenith transmission $T_{\rm atm}(\lambda)$ comes either from a site-specific FITS table (e.g. the Paranal product, whose `EXTINCTION` column in mag airmass^{-1} is converted once as $T = 10^{-0.4k_\lambda}$) or from the built-in broad-band U–K table (Table 2). The Bouguer law applies it as T^X, exactly once, to the source; observed sky brightnesses are never extinguished a second time.

5.2 Telluric molecular bands

A smooth extinction curve describes Rayleigh scattering, aerosols and ozone, but not the discrete molecular absorption *bands* of O_2 and H_2O that structure the red and near-infrared: the O_2 B and A bands at 6870 and 7605 Å (the A band removes up to $\sim$75% of the light at its core), and the water bands near 7200, 8200, 8900 and 9400 Å. As an option, SPETC multiplies the smooth transmission by a parametric band model: each band is a Gaussian absorption profile of tabulated centre λ_0, width σ_b and zenith depth d_0, with the depth growing with airmass as $d(X) = d_0\, X^p$. The exponent p encodes the curve of growth — the regime describing how absorption grows with absorber amount: $p = 1$ for the optically thin water lines (absorption $\propto$ absorbers) and $p \simeq 0.55$ for the saturated O_2 bands, whose cores are already black so only the band *wings* deepen ($\propto \sqrt{X}$). This is a band-averaged model adequate for S/N budgeting at the ETC's resolution; it is not a line-by-line radiative transfer, and it deliberately omits the strong variability of the water columns, which only a contemporaneous measured spectrum can capture.

5.3 Rise, set and twilight conventions

Positions in the planning track are *apparent*: they include atmospheric refraction (evaluated with ERFA at the site conditions), which lifts objects near the horizon by up to $\sim 34'$. Two published conventions are nevertheless defined on different quantities, and SPETC follows both correctly (v10.2). Twilight boundaries — civil ($-6°$), nautical ($-12°$) and astronomical ($-18°$) — are defined on the *geometric* (unrefracted) altitude of the Sun's centre, so the sky model receives the geometric solar altitude, computed in a refraction-free frame. Sunrise and sunset are the almanac convention: the Sun's *upper limb* touching the geometric horizon under standard refraction, i.e. the centre at geometric altitude $-50' = -(34' + 16')$, refraction plus solar semidiameter.

5.4 Refraction and airmass

Apparent altitudes include ERFA atmospheric refraction evaluated at the ISA pressure and temperature of the site elevation, $P = 1013.25\,\mathrm{hPa}\, e^{-E/8435\,\mathrm{m}}$ and $T = 15\,°\mathrm{C} - 0.0065\,E$. The airmass is the Pickering [2002] interpolative formula on the *apparent* altitude h (degrees),

$$X = \frac{1}{\sin\big(h + 244/(165 + 47\,h^{1.1})\big)}, \tag{5}$$

which matches sec z near the zenith and remains physical to the horizon, where the plane-parallel form diverges (Fig. 1). The ETC deliberately reports no airmass below $h = 5°$.

5.5 Differential atmospheric refraction

The refractive index of air follows Filippenko [1982]: at standard conditions

$$(n-1)\,10^6 = 64.328 + \frac{29498.1}{146 - \sigma^2} + \frac{255.4}{41 - \sigma^2}, \qquad \sigma = 1/\lambda_{\mu m}, \tag{6}$$

rescaled to the site pressure and temperature with his equations (2)–(3), including the water-vapour term. The differential refraction relative to a reference wavelength is

$$\Delta R(\lambda) = 206265\,\big[n(\lambda) - n(\lambda_{\rm ref})\big]\,\tan z, \tag{7}$$

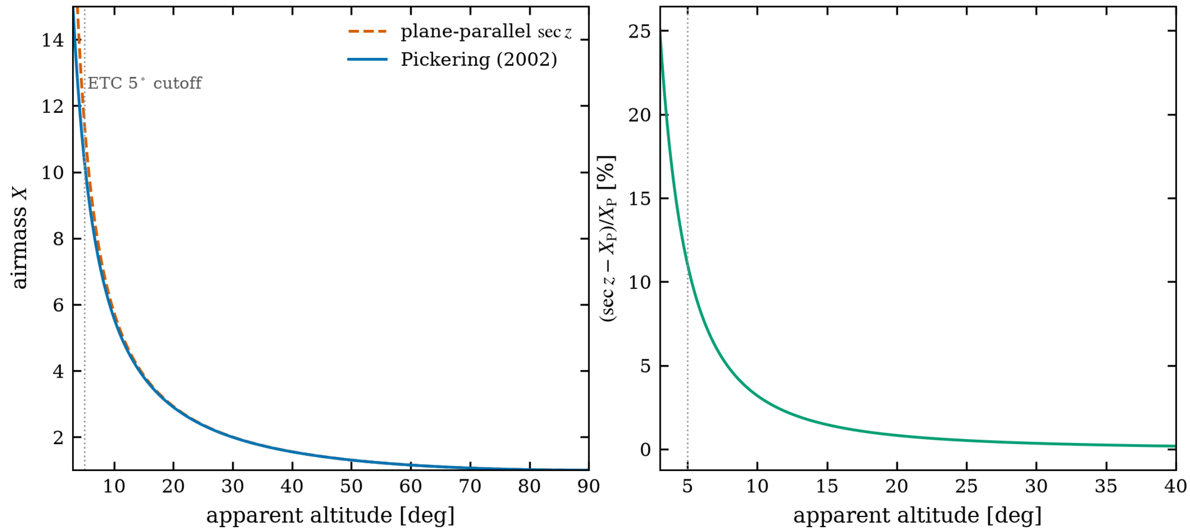

Figure 1: Left: Pickering (2002) airmass versus plane-parallel sec z. Right: relative overestimate of sec z; at the ETC's $5°$ cutoff it already exceeds 10%.

about $1''$ between 4000 and 5500 Å at $z = 45°$ (Fig. 2). Its use in slit losses is described in Sect. 9.2; the parallactic angle

$$q = \arctan \frac{\sin H}{\tan \varphi \cos \delta - \sin \delta \cos H} \tag{8}$$

(H hour angle, φ latitude, δ declination) is reported for every time sample, because a slit oriented at q suffers no dispersion loss.

5.6 The local terrain horizon

The astronomical horizon of a real site is set by terrain, not by altitude zero. SPETC computes the apparent horizon profile $\theta_h(A)$ from the Copernicus GLO-30 global digital surface model (30-m resolution): the elevation field within a user-chosen radius r (1–100 km) is resampled onto a local East/North metric grid, and along each of 360 azimuth rays the maximum apparent elevation angle is taken, where the apparent angle of a terrain point at distance d and height difference Δh is

$$\theta = \arctan \frac{\Delta h - d^2/2R_\oplus}{d} + 34', \tag{9}$$

combining the Earth-curvature drop $d^2/2R_\oplus$ with the standard $34'$ horizontal refraction (the same convention used for rise/set phenomena). The choice of r is a statement about what can block the site: 10 km captures the surrounding hills of a typical site; a plain below a mountain chain may need 50–100 km before the profile converges. The profile is stored as a per-degree CSV; a natural future use is masking the visibility track and the S/N time series where the target sits below the *local* rather than the geometric horizon.

5.7 Scintillation

Intensity scintillation follows the Young [1967] law, kept in the functional form endorsed by Osborn et al. [2015],

$$\frac{\sigma_{\text{scint}}}{I} = 0.09\, d_{\text{cm}}^{-2/3}\, X^{1.75}\, e^{-E/8000\,\text{m}}\, (2\, t_{\text{exp}})^{-1/2}, \tag{10}$$

with d the aperture diameter, E the site elevation and t_{exp} the exposure time. Because this noise is multiplicative in the source flux, it dominates bright-star photometry and imposes a hard S/N ceiling $\simeq I/\sigma_{\text{scint}}$ per exposure (Fig. 6).

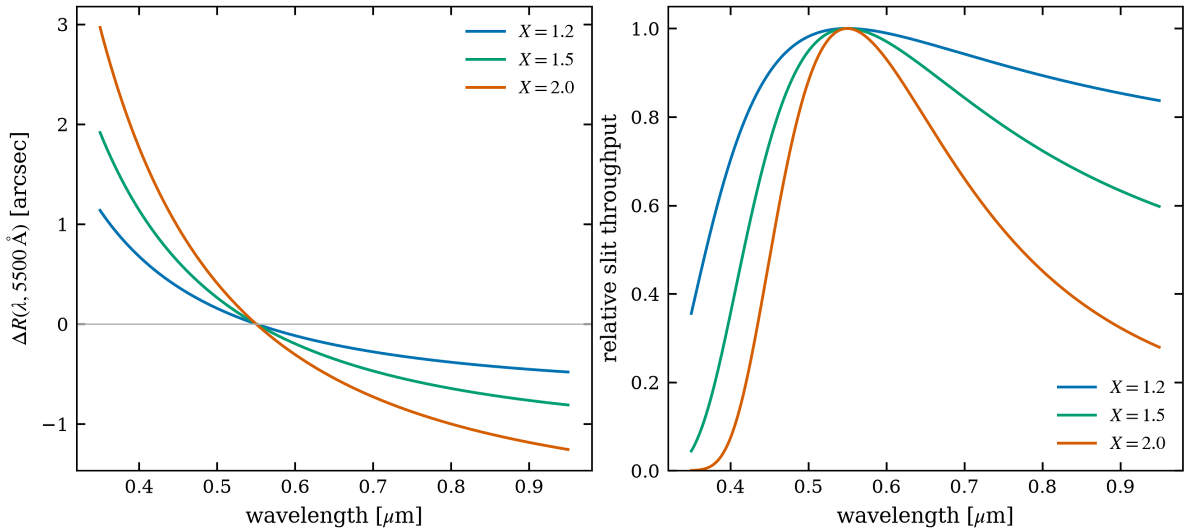

Figure 2: Left: differential refraction relative to 5500 Å for three airmasses. Right: resulting relative slit throughput for a $1.5''$ slit in $1.5''$ seeing when the slit is *not* at the parallactic angle (worst-case geometry).

The scalings deserve a moment's reflection, because they run against photon intuition. Photon noise improves with a brighter star; scintillation does not — the fractional modulation is a property of the turbulence, not of the source, so above the crossover magnitude (roughly $V \sim$ 10–12 for decimetre apertures at 60 s) nothing about the star helps any more. What does help: aperture ($d^{-2/3}$ — a 40-cm telescope has $1.6\times$ less scintillation noise than a 20-cm), airmass (a steep $X^{1.75}$ — observing at $X = 2$ instead of 1.2 doubles it), altitude of the *site* (the $e^{-E/8\,\mathrm{km}}$ factor is why mountain observatories twinkle less), and total time ($t^{-1/2}$, whether in one frame or many — Sect. 8.2). For two stars measured differentially the correlation of their scintillation decays with angular separation; SPETC conservatively treats the two as uncorrelated (Sect. 8.3), which for arcminute separations is close to the truth at short exposures.

6 Night-, twilight- and day-sky model

For faint targets the sky, not the detector, is the adversary: once the background electrons in the aperture outnumber the source electrons, the S/N grows only as the square root of exposure time and every magnitude of extra sky brightness costs a factor 2.5 in required time. An ETC is therefore only as honest as its sky model. SPETC's model has an unusual property for a tool in this class: it is *position- and time-dependent*. The same target yields a different sky — and a different exposure recommendation — depending on where it sits with respect to the ecliptic (zodiacal light), the Milky Way (integrated starlight), the Moon (scattered moonlight), the Sun (twilight), and the local zenith (airglow path length), all evaluated per time sample of the night. The components and their assembly are as follows.

6.1 Dark sky

Surface brightnesses are budgeted in *S10 units* — the brightness equivalent to one star of visual magnitude 10 per square degree, a linear (flux-like) unit in which components simply add; the conversion to magnitudes per square arcsecond is $\mu = 27.78 - 2.5\log_{10} q$ for a brightness of q S10. The dark-sky normalization is the broad-band U–K surface-brightness table of Table 2 (from the ING model of Benn and Ellison 1998), corrected for the pointing and epoch through the S10 component budget

$$q_3 \;=\; q_{\rm air}\, V(z)\, \mathcal{E}(z) \;+\; q_{\rm zod}(|\beta|)\, g(\epsilon) \;+\; q_*(|b|), \tag{11}$$

normalized to the dark reference composition $q_{\rm ref} = 145 + 60$ S10. The components are:

Table 2: Built-in broad-band sky data (U–K): dark-sky surface brightness, extinction coefficient, Vega zero point and Krisciunas–Schaefer lunar colour factor.

Band	λ [Å]	$\mu_{\rm dark}$ [mag/$''^2$]	k [mag/X]	$Z_\nu^{\rm Vega}$ [Jy]	lunar factor
U	3650	22.0	0.550	1810	2.00
B	4450	22.7	0.250	4260	1.60
V	5510	21.9	0.150	3640	2.40
R	6580	21.0	0.090	3080	3.00
I	8060	20.0	0.060	2550	5.90
Z	10000	18.8	0.050	2250	8.94
J	12500	16.1	0.100	1594	13.06
H	16500	14.7	0.110	1024	18.09
K	22000	12.5	0.070	667	24.04

- **Airglow**: $q_{\rm air} = 145 + 130\,(a - 0.8)/1.2$ S10, where the solar-cycle activity $a \in [0.8, 2.0]$ is interpolated between the tabulated solar minima (airglow is excited by solar UV, so it roughly doubles from solar minimum to maximum). The line-of-sight enhancement is the van Rhijn path-length law [van Rhijn, 1921] for an emitting layer at height $\hat{h} = 90$ km,

  $$V(z) = \left[1 - \left(\frac{R_\oplus}{R_\oplus + \hat{h}}\sin z\right)^2\right]^{-1/2}, \tag{12}$$

  which is sub-linear in airmass and bounded ($V \simeq 6$) at the horizon (Fig. 3, right). The raw van Rhijn factor is however never observed: the emitted light must still descend through the troposphere along the same slant, which attenuates it by

  $$\mathcal{E}(z) \;=\; 10^{-0.4\,k_V\,[X_{\rm KY}(z)-1]}, \tag{13}$$

  with k_V the V extinction coefficient and $X_{\rm KY}$ the Kasten and Young [1989] line-of-sight airmass $[\cos z + 0.50572\,(96.07995 - z)^{-1.6364}]^{-1}$, finite ($X \simeq 38$) at the horizon. Near the horizon $\mathcal{E}$ partially cancels the van Rhijn enhancement, matching the observed behaviour (v10.2).

- **Zodiacal light** — sunlight scattered by interplanetary dust: $q_{\rm zod} = 140 - 90\sin|\beta|$ S10 for $|\beta| < 60°$, else 60 S10, symmetric about the ecliptic. This latitude law describes the anti-solar sky; towards the Sun the dust column and the scattering efficiency both rise steeply. The solar elongation ϵ (the field's angular distance from the Sun) therefore multiplies it by the factor

  $$g(\epsilon) \;=\; \begin{cases} (\epsilon/110°)^{-2.3} & 30° \le \epsilon < 110° \\ 1 & \epsilon \ge 110°, \end{cases} \tag{14}$$

  a power-law fit to the Leinert et al. [1998] tables at $\beta = 0$ ($\sim 4\times$ brighter at $\epsilon = 60°$); below 30° the line of sight is unobservable and the factor is held (v10.2).

- **Integrated starlight** — the summed light of unresolved stars: $q_* = 100\,e^{-|b|/10°}$ S10, falling away from the galactic plane.

The field's ecliptic and galactic latitudes are computed from the entered ICRS coordinates with `astropy`; the sky is therefore genuinely position-dependent (Fig. 3, left).

6.2 Moonlight

Moonlight follows Krisciunas and Schaefer [1991]. The lunar phase angle α is the Sun–Moon–Earth angle (0° = full Moon, 90° = quarter), computed as 180° minus the Sun–Moon elongation. The lunar illuminance — the flux of moonlight arriving at the top of the atmosphere — is $I^* = 10^{-0.4(3.84 + 0.026|\alpha| + 4\times10^{-9}\alpha^4)}\,(\bar{d}/d_{\rm moon})^2$:

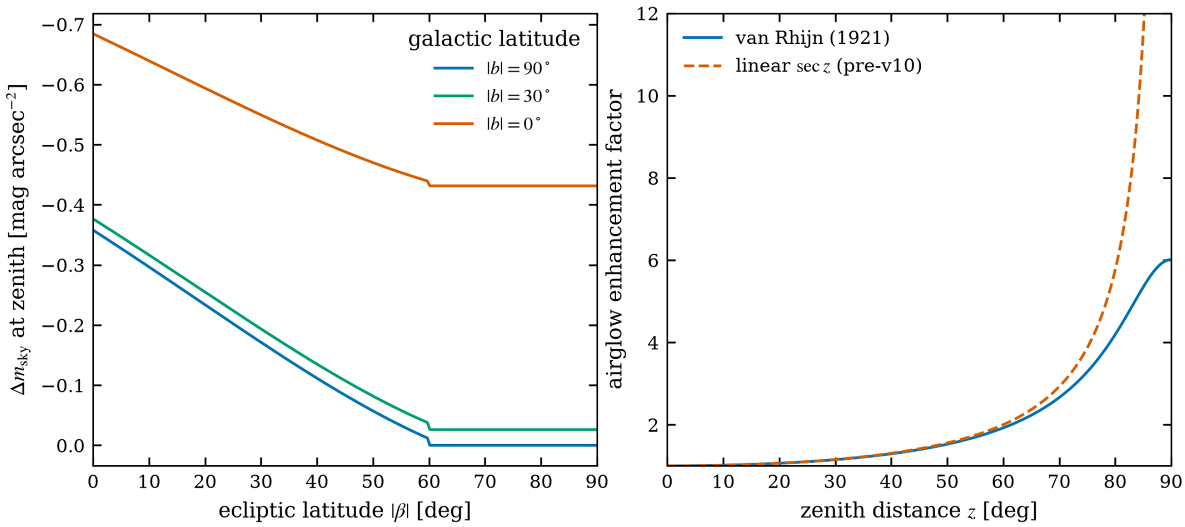

Figure 3: Left: zenith dark-sky brightening relative to the tabulated values as a function of ecliptic latitude, for three galactic latitudes (solar minimum). Right: van Rhijn airglow enhancement versus the linear sec z scaling used before v10.

the polynomial is K&S's fit to the lunar phase curve, and the second factor is the inverse-square modulation by the topocentric Moon distance $d_{\rm moon}$ around its mean $\bar{d} = 384\,400$ km — a $\pm 15\%$ brightness swing over the anomalistic month that the track supplies per time sample (v10.2). The scattering function at Moon–target separation ρ is

$$f(\rho) \;=\; 10^{5.36}\big(1.06 + \cos^2\rho\big) \;+\; 10^{6.15-\rho/40^\circ}. \tag{15}$$

The scattering function has two physically distinct terms: $10^{5.36}(1.06+\cos^2\rho)$ is the Rayleigh contribution (molecular scattering, with its characteristic $1+\cos^2$ angular pattern), and $10^{6.15-\rho/40^\circ}$ is the aerosol *aureole*, the forward-scattering halo around the Moon. The moonlit surface brightness in nanoLambert (a luminance unit; 1 nL converts to 3.8 S10) is $B_{\rm moon} = I^* f(\rho)\, 10^{-0.4kX_{\rm m}}\big(1 - 10^{-0.4kX}\big)$, where $X_{\rm m}$ is the Moon's airmass (the moonlight is first extinguished on its way down) and the bracket is the fraction of light scattered *out of* the direct beam along the target's path — the scattered moonlight the target line of sight picks up. This is converted to S10 and distributed over the nine bands with the lunar colour factors of Table 2. Equation (15) corrects a long-standing transcription error in which the constant $10^{5.36}$ appeared inside the exponent; the two forms diverge by more than three orders of magnitude within 30° of the Moon (Fig. 4).

6.3 SQM and Bortle input

Amateur observers know their sky as a Sky Quality Meter reading (zenith V surface brightness) or a Bortle [2001] class rather than as a per-band table. In `sqm` mode the entered zenith SQM value $\mu_{\rm SQM}$ replaces the dark-sky normalization directly:

$$\mu(\lambda_{\rm piv}) \;=\; \mu_{\rm SQM} \;+\; \big[\mu_{\rm tab}(\lambda_{\rm piv}) - \mu_{\rm tab}(V)\big], \tag{16}$$

i.e. the built-in table supplies only the band colour, while the SQM reading — which already contains the site's light pollution and current airglow — sets the level. The Krisciunas–Schaefer Moon contribution of the previous subsection is applied on top unchanged. A Bortle-class selector fills typical zenith SQM values (21.9, 21.8, 21.6, 21.1, 20.5, 19.8, 19.0, 18.0, 17.0 mag arcsec^{-2} for classes 1–9).

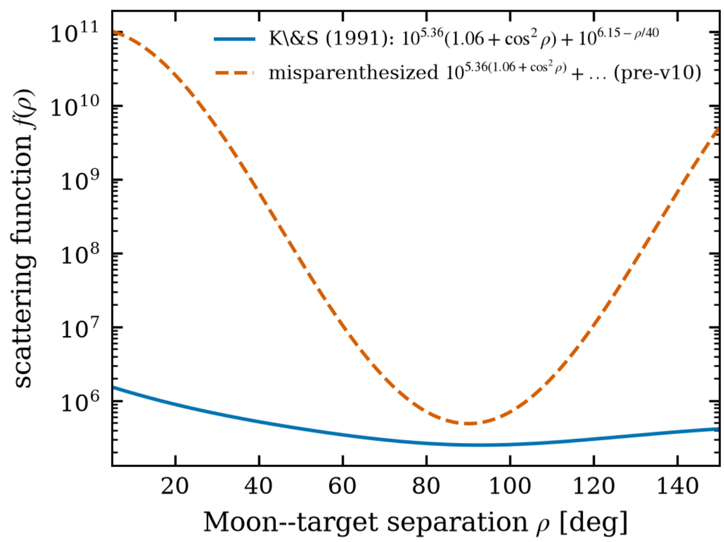

Figure 4: The Krisciunas and Schaefer [1991] scattering function (solid) and the misparenthesized variant present before v10 (dashed).

6.4 Twilight and daylight

Between *geometric* Sun altitudes of -18° (end of astronomical twilight) and 0° the dark and daylight skies are blended linearly in flux — magnitudes cannot be averaged, fluxes can. Above the horizon a daylight table in the Weaver [1947] lineage supplies the V-band surface brightness as a function of solar and sky zenith distance, mapped to other bands with the dark-sky colours of Table 2.

The v10.1 physics audit found the daylight table, read directly as mag/arcsec^2, to be $\sim$4.5 magnitudes too *faint*: its values (8.5–9.2) would make the daytime sky only $\sim$200$\times$ brighter than a dark night, whereas the physical clear daytime zenith sky is $\sim$3000–8000 cd/m^2, i.e. $V \simeq 3.5$–4.5 mag/arcsec^2 against the night-sky anchor (21.9 mag/arcsec$^2 \approx 2 \times 10^{-4}$ cd/m^2) — a $\sim$$10^7$ flux ratio. The v10.2 model therefore keeps the table's *shape* in solar and sky zenith distance but re-anchors its brightest entry to the physical $V = 4.0$ mag/arcsec^2; the twilight blend inherits the corrected anchor. Both regimes remain broad-band approximations, clearly flagged in the user interface, and the original supplied table is preserved in the source for the day the exact Weaver unit conversion is re-derived.

7 Point-spread function and aperture coupling

The point-spread function is the currency in which three different questions are paid: how much of a star's light enters a photometric aperture of a given radius, how much passes a slit of a given width and decentring, and how much lands on the single brightest pixel (the saturation question). All three are integrals of the same profile over different domains, and all three are sensitive to the profile's *wings*: real seeing profiles carry far more energy at large radii than a Gaussian of the same FWHM, which makes Gaussian-based aperture corrections optimistic and Gaussian-based saturation predictions slightly pessimistic. SPETC therefore offers both the Gaussian (for comparability with other tools) and the Moffat profile (for realism), everywhere through analytic expressions — no PSF is ever rendered or numerically integrated, which keeps the whole-night time series fast.

The seeing-limited PSF is selectable between a circular Gaussian and a circular Moffat [1969] profile

$$I(r) \propto \left(1 + \frac{r^2}{\alpha^2}\right)^{-\beta}, \qquad \alpha = \frac{\mathrm{FWHM}}{2\sqrt{2^{1/\beta} - 1}}, \tag{17}$$

whose power-law wings ($\beta \simeq 2.5$–4.7; the Gaussian is the $\beta \to \infty$ limit) reproduce real seeing profiles that a Gaussian underestimates.

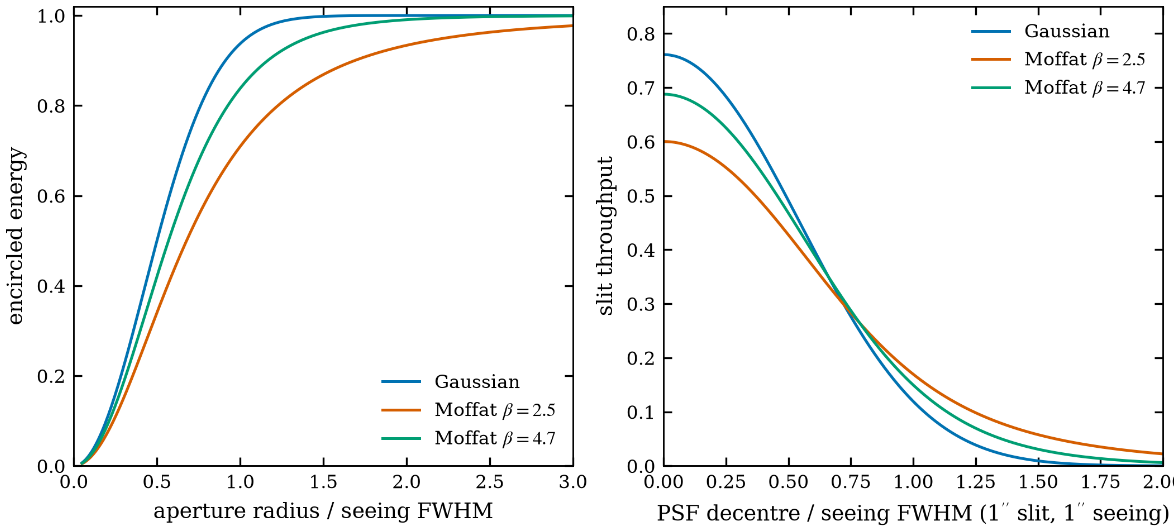

Figure 5: Left: encircled energy versus aperture radius in units of the seeing FWHM. Right: slit throughput of a slit of one seeing-FWHM width as the PSF is decentred, Eq. (19); the Moffat wings both lower the centred coupling and soften the decentring penalty.

Encircled energy. For a circular aperture of radius r,

$$\mathrm{EE}_{\mathrm{Gauss}}(r) = 1 - e^{-r^2/2\sigma^2}, \qquad \mathrm{EE}_{\mathrm{Moffat}}(r) = 1 - \left(1 + \frac{r^2}{\alpha^2}\right)^{1-\beta}. \tag{18}$$

Slit coupling. The flux through an infinite slit of width w whose centre is displaced by δ from the PSF centroid uses the one-dimensional marginal profile. For the Gaussian this is the usual error function; for the circular Moffat the marginal is *exactly* a Student-t distribution with $\nu = 2\beta - 2$ degrees of freedom and scale $\alpha/\sqrt{\nu}$, so

$$T(w, \delta) \;=\; F_\nu\!\left(\frac{w/2 - \delta}{\alpha/\sqrt{\nu}}\right) - F_\nu\!\left(\frac{-w/2 - \delta}{\alpha/\sqrt{\nu}}\right), \tag{19}$$

with F_ν the Student-t CDF — no numerical integration is required. The same expression with w equal to one pixel gives the central-pixel fraction used for saturation prediction, as the separable product of the dispersion and spatial couplings. Figure 5 compares the models.

7.1 Wavelength and airmass dependence of the seeing

The seeing FWHM is not a constant of the night: Kolmogorov turbulence gives a Fried parameter $r_0 \propto \lambda^{6/5}(\cos z)^{3/5}$ and hence a seeing FWHM $\propto \lambda/r_0$,

$$\mathrm{FWHM}(\lambda, X) \;=\; s_{V,\mathrm{zenith}}\; X^{0.6} \left(\frac{\lambda}{5000\,\text{Å}}\right)^{-0.2}, \tag{20}$$

where $s_{V,\mathrm{zenith}}$ is the entered zenith seeing in V and X the airmass. This is the exact form used by ETC-42 [CeSAM/LAM, Surace and CeSAM/LAM, 2013, `calculator/psf/Seeing.java`]. As an optional mode, SPETC applies Eq. (20) per wavelength in the spectroscopic slit and extraction couplings (so blue light suffers larger losses than red through the same slit), and at the observing band's pivot wavelength in photometry. Because the airmass factor is evaluated per time sample, the whole-night time series then correctly shows the seeing — and the losses that follow from it — degrading as the target sets. The mode is off by default, leaving the seeing flat; when it is on, the entered value is by definition the zenith V seeing.

7.2 Guiding blur

Over a long exposure the star also moves: periodic error, polar-alignment drift and guiding corrections smear the image. If the tracking error is approximately Gaussian with rms σ_g per axis, the time-averaged image is the seeing profile convolved with a Gaussian of FWHM $2.355\,\sigma_g$, and the effective image width entering every coupling of this section is

$$\mathrm{FWHM}_{\mathrm{eff}} \;=\; \sqrt{\mathrm{FWHM}^2_{\mathrm{seeing}} + (2.355\,\sigma_g)^2}. \tag{21}$$

An entered guiding rms of 0 (the default) reproduces the pure-seeing case; a typical unguided minute on a mid-range mount ($\sigma_g \sim 1''$) visibly lowers aperture and slit couplings, which is exactly the effect amateurs observe between guided and unguided frames (v10.2).

7.3 Defocused PSF: the geometric donut

Defocused photometry spreads a bright star deliberately over many pixels to suppress scintillation, flat-field residuals and saturation. When the detector is displaced by a distance δ from the focal plane, the geometric image of a point source is the projection of the (annular) pupil: a uniformly-illuminated ring, or "donut". For a primary of diameter D and focal length F (focal ratio $N = F/D$), similar triangles give a blur circle of full diameter $\delta/N = \delta D/F$, i.e. an outer radius on the focal plane

$$r_{\mathrm{out}} \;=\; \frac{\delta D}{2F}, \tag{22}$$

which depends only on $|\delta|$: intra- and extra-focal displacements of equal magnitude give identical images. A central obstruction of linear fraction $\varepsilon = D_{\mathrm{obs}}/D$ (a classical Ritchey–Chrétien or Cassegrain, $\varepsilon > 0$) blocks the inner cone and leaves a hole of radius $r_{\mathrm{in}} = \varepsilon\, r_{\mathrm{out}}$; a classical reflector modelled with $\varepsilon = 0$ gives a filled disc. In the pure geometric limit a uniformly illuminated pupil maps to a *flat* annulus, whose encircled energy would be the analytic

$$\mathrm{EE}_{\mathrm{geo}}(r) \;=\; \begin{cases} 0, & r < r_{\mathrm{in}}, \\ \dfrac{r^2 - r^2_{\mathrm{in}}}{r^2_{\mathrm{out}} - r^2_{\mathrm{in}}}, & r_{\mathrm{in}} \le r \le r_{\mathrm{out}}, \\ 1, & r > r_{\mathrm{out}}. \end{cases} \tag{23}$$

That flat annulus, with its unphysical vertical edges, is only the *ideal* limit — it is not what a camera records. The image the detector actually sees is the geometric annulus **convolved with the atmospheric point-spread function**,

$$I(\mathbf{r}) \;=\; A(\mathbf{r}) \,*\, \mathrm{PSF}_{\mathrm{seeing}}(\mathbf{r}), \tag{24}$$

where A is the annulus indicator and $\mathrm{PSF}_{\mathrm{seeing}}$ is the selected Gaussian or Moffat profile with any guiding rms added in quadrature to its FWHM. The program builds A on a two-dimensional grid, convolves it with the seeing kernel by FFT, and azimuthally averages the result; the encircled energy $\mathrm{EE}(r)$ then follows by cumulative radial integration of I. The convolution rounds the rims over roughly one seeing width, partly fills the central hole and adds the wings a real defocused star shows — so a small defocus (donut comparable to or below the seeing) correctly collapses to a merely broadened stellar core, while a large defocus keeps a clear ring.

Focal-plane radii convert to angle through the plate scale ($1\,\mu\mathrm{m} \to 206264.8 \times 10^{-3}/F_{\mathrm{mm}}$ arcsec) and to pixels through the pixel pitch. The user selects a photometry aperture radius r_{ap} from the encircled-energy curve; the captured light fraction is $\mathrm{EE}(r_{\mathrm{ap}})$, and the peak pixel holds the central convolved surface brightness (normalised to unit total flux) times the pixel solid angle — orders of magnitude below the focused stellar peak, which is the whole benefit of defocusing.

The remaining approximations are diffraction ringing at the edges and optical aberrations (spherical, coma; small on-axis for a paraboloid or an RC), both sub-dominant once the donut is much larger than the seeing and diffraction scales — the regime in which defocused photometry operates.

Table 3: Noise terms and their scalings.

Term	Variance [e^{-2}]	Scaling	Reference
Source shot	S	$\propto t$	—
Sky shot	B	$\propto t\, r^2$	Sect. 6
Dark	D	$\propto t\, N_{\text{pix}}$	—
Read	$N_{\text{pix}}\sigma_R^2$	per frame	—
Scintillation	$(S\,\sigma_{\text{scint}}/I)^2$	$\propto t^{1\ a}$	Young [1967], Osborn et al. [2015]
Quantization	$N_{\text{pix}}\, g^2/12$	per frame	Janesick [2001]
Extra background	$\dot{b}\, N_{\text{pix}}\, t$	$\propto t\, N_{\text{pix}}$	Surace and CeSAM/LAM [2013]
Sky subtraction	$\times\,(1 + N_{\text{pix}}/n_{\text{sky}})$	on background terms	Merline and Howell [1995]

The *extra background* $\dot{b}$ [e$^-$/s/pixel] is an optional catch-all Poisson term for detector glow, stray/scattered light or ghosting not covered by the sky model, adopted from ETC-42's `ExtraBackgroundNoise`; it also enters the peak-pixel prediction.

[a] $S \propto t$ while $\sigma_{\text{scint}}/I \propto t^{-1/2}$, so the S/N ceiling $I/\sigma_{\text{scint}} \propto \sqrt{t}$.

8 Photometric ETC

Aperture photometry is conceptually the simplest measurement the program models — add up the electrons in a circle, subtract the sky — and precisely for that reason it is where every term of the noise budget can be seen individually at work. This section assembles the budget term by term, then treats the three questions layered on top of it: when does the detector saturate; how should a long integration be split into frames; and what precision does a *differential* measurement against a comparison star achieve, which is the number variable-star and exoplanet observers actually need. A reader who wants one formula to take away should take Eq. (25); a reader who wants one insight should take Fig. 6: from the ground, bright-star photometry is scintillation-limited, faint-star photometry is sky-limited, and the detector matters mostly in the narrow corridor between the two.

8.1 Signal and noise budget

For a circular aperture of radius r containing $N_{\text{pix}} = \pi r^2/p^2$ pixels (p = plate scale per pixel, arcsec), the extracted source signal is $S = \dot{S}\,\mathrm{EE}(r)\,t_{\text{exp}}$ electrons (rate × encircled-energy fraction × exposure), the sky signal is $B = \dot{B}\,\pi r^2\,t_{\text{exp}}$ from the observed surface brightness of Sect. 6 ($\dot{B}$ in e$^-$/s/arcsec^2), and the dark charge is $D = \dot{d}\,N_{\text{pix}}\,t_{\text{exp}}$ with $\dot{d}$ the dark-current rate per pixel. The S/N is the CCD equation extended with the multiplicative, quantization and background-estimation terms:

$$\mathrm{S/N} \;=\; \frac{S}{\sqrt{S + f_{\text{sub}}\left[B + D + N_{\text{pix}}\,\sigma_R^2 + N_{\text{pix}}\,g^2/12\right] + \left(S\,\sigma_{\text{scint}}/I\right)^2}}, \qquad f_{\text{sub}} = 1 + \frac{N_{\text{pix}}}{n_{\text{sky}}}, \quad (25)$$

with the scintillation fraction of Eq. (10), the ADC quantization variance $g^2/12$ per pixel [Janesick, 2001], and the sky-subtraction factor f_{sub} [Merline and Howell, 1995]: the classical CCD equation assumes the per-pixel background is known exactly, but in practice it is *estimated* from n_{sky} annulus pixels and subtracted, which propagates the estimate's own noise into every aperture pixel. For $n_{\text{sky}} = 3N_{\text{pix}}$ the background noise grows by 15%; $n_{\text{sky}} \to \infty$ (or the entered 0) recovers the ideal equation. Scintillation stays outside the factor because it is carried by the source, not the background. Flat-field residuals are deliberately not modelled. The inverse problem — the exposure that reaches a requested S/N — is solved in closed form from the quadratic obtained from Eq. (25) with the Poisson terms proportional to t.

8.2 Exposure solving and stack planning

The exposure reaching a requested S/N solves the quadratic obtained from Eq. (25). Crucially, the scintillation variance is *linear* in time — with $\sigma/I = C(2t)^{-1/2}$ the variance of $\dot{S}t$ electrons is $(\dot{S}C)^2 t/2$

— so it enters the closed form exactly as an additional Poisson-like rate added to $Q = \dot{S} + \dot{B} + \dot{D}$:

$$t \;=\; \frac{\mathrm{SNR}^2\big(Q + \dot{S}^2C^2/2\big) + \sqrt{\cdot\;^{2} + 4\dot{S}^2\,\mathrm{SNR}^2 N_{\mathrm{pix}}\sigma_R^2}}{2\dot{S}^2}. \tag{26}$$

Without this term a bright-star exposure request undershoots badly (a demonstration case returned S/N 230 for a 500 target); with it the solved exposure reproduces the requested S/N to better than 2% (regression tested).

The **stack planner** answers the practical amateur question "how many subs?": the sub-exposure is the shortest of the user's preference and the saturation limit of the brightest pixel; the total time for N frames of length t_{sub} follows in closed form from the stacked CCD equation $\mathrm{SNR} = \dot{S}T/\sqrt{QT + NN_{\mathrm{pix}}\sigma_R^2}$ (the scintillation rate is framing-independent, since it averages down with total time). Reported alongside are the read-noise penalty relative to one ideal long exposure and the frame length above which the background variance exceeds $10\,\sigma_R^2$ per pixel — the classical "sky-limited" criterion.

8.3 Differential photometry

For variable-star and exoplanet work the scientific quantity is the differential precision against a comparison star of magnitude m_c, in millimagnitudes per frame:

$$\sigma_{\mathrm{diff}} \;=\; 1085.7\,\sqrt{\mathrm{SNR}_t^{-2} + \mathrm{SNR}_c^{-2}} \quad [\mathrm{mmag}], \tag{27}$$

where both S/N values carry the full budget of Eq. (25) — each star scintillates — and the two scintillation terms are treated as uncorrelated, which is conservative for the arcminute-scale separations typical of ensemble photometry [Osborn et al., 2015].

8.4 CMOS gain description

Amateur CMOS cameras change conversion gain, read noise and full well *together* with the gain setting. The detector is therefore describable by a per-camera table (setting $\to$ e$^-$/ADU, σ_R, FWC) rather than one fixed triple; the selected row feeds Eq. (25), the quantization term $g^2/12$ and the saturation limits simultaneously, so switching the gain setting consistently moves the S/N, the ADC ceiling and the maximum unsaturated exposure.

8.5 One-shot-colour (Bayer) sensors

For a colour camera read out as a single channel, the Bayer mosaic assigns the channel a fixed area fraction f of the focal plane ($f = 1/2$ for green, $1/4$ for red and blue). Aperture- or extraction-integrated source and sky rates therefore scale by f, and only $f N_{\mathrm{pix}}$ of the aperture pixels contribute read and dark noise for that channel. The *peak pixel* is not scaled: a channel pixel at the PSF centre receives the full local flux through its own dye, so saturation prediction is unchanged. The quantum-efficiency curve must be the channel's *effective* QE — sensor QE times the colour-filter-array dye transmission — which manufacturers publish per channel. Without this treatment an OSC user overestimates the collected signal by 2–4×.

8.6 Saturation

Saturation is evaluated on the brightest pixel of the *unextracted* image: the PSF central-pixel fraction (Sect. 7) times the total source charge, plus the per-pixel sky and dark. The peak is compared with both the full well and the ADC ceiling $g\,(2^{N_{\mathrm{bit}}} - 1)$, and the flag distinguishes `FULL_WELL`, `ADC`, `BOTH` and `NONE`; the maximum unsaturated single-frame exposure is reported at every time sample.

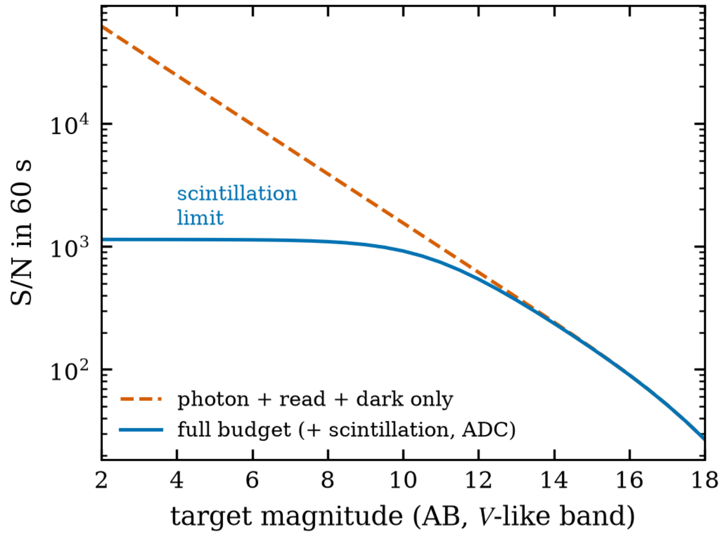

Figure 6: Broad-band S/N in 60 s for the 358-mm reference configuration, computed by the released code. The full budget (solid) saturates at the scintillation ceiling $\simeq 1.2 \times 10^3$; a Poisson-only model (dashed) overestimates bright-star performance by more than an order of magnitude.

8.7 Extended sources

Galaxies, nebulae and planets are modelled as uniform surface brightness over a stated angular area Ω: the entered magnitude remains the *integrated* magnitude, so the mean surface rate is $\dot{S}/\Omega$ per arcsec^2. The aperture then captures the area fraction $\min(\pi r^2/\Omega,\, 1)$ with no PSF concentration loss, and the peak pixel holds $\dot{S}\, p^2/\Omega$ — the appropriate saturation criterion for a resolved source. The approximation is stated to be valid when the source is much larger than the seeing disc, where PSF blur merely redistributes light across the source edge.

9 Spectroscopic ETC

Spectroscopy divides the same photons that photometry collects into hundreds or thousands of wavelength bins, so its S/N per bin is lower by $\sqrt{N_{\text{bins}}}$-like factors and every loss mechanism bites harder. Two geometries are modelled, and they fail in opposite ways. A *slit* spectrograph defines its resolution optically and pays for it at the entrance: slit losses, extraction losses, and — unless the slit tracks the parallactic angle — chromatic losses from atmospheric dispersion that can silently remove half the blue signal. A *slitless* grating (the Star Analyser class) pays nothing at an entrance it does not have, collects every photon of source *and* sky at every wavelength, and instead surrenders resolution to the seeing: its resolution element is the star image itself, dispersed. The engine treats both with the same per-element bookkeeping so that their trade-off can be compared honestly for the same hardware.

All spectroscopic quantities are per resolution element ("resel"): the wavelength interval $\Delta\lambda$ the instrument cannot subdivide, one output row of the ETC. The wavelength grid is logarithmic at constant R in slit mode, or linear with the resolution element $\Delta\lambda = \mathcal{D}\,\ell$ in slitless mode ($\mathcal{D}$ = dispersion in Å/pixel, ℓ = LSF FWHM in pixels). Source and sky rates follow Eq. (4) restricted to each element, and the S/N applies Eq. (25) with N_{pix} = (dispersion pixels per element) $\times$ (spatial extraction pixels).

9.1 Per-element integration: why bin integrals, not samples

How the template flux is booked into resolution elements decides whether line spectroscopy is trustworthy. Sampling the template at the element's central wavelength and multiplying by $\Delta\lambda$ — the obvious shortcut — is correct only for spectra smooth on the scale of $\Delta\lambda$. For a spectral *line* narrower than the element it is wrong in a way that depends on luck: the line is fully counted if a sample lands on it and lost if none does, so the *total* detected line flux changes with the chosen R — unphysical, since dispersing the same photons more or less finely cannot create or destroy them. The v10.1 audit measured an $8\times$ spread in the detected counts of a 2 Å emission line between $R = 200$ and $R = 20000$ under point sampling. Since v10.2 each element's source counts are the exact *integral* of the calibrated template over the element,

$$\dot{S}_i \;=\; \eta\, A \int_{\lambda_i-\Delta\lambda_i/2}^{\lambda_i+\Delta\lambda_i/2} F_\lambda^{\rm cal}\, T_\lambda\, Q_\lambda\, O_\lambda\, T_{\rm atm}^X\, \frac{\lambda}{hc}\, d\lambda, \tag{28}$$

evaluated for all elements at once by cumulative trapezoidal integration on the union of the template grid and the element edges (exact for the piecewise-linear tabulated inputs, and $O(n)$ — a single pass even at $R = 10^5$). Total line counts are now invariant under R to better than 0.2% (regression-tested), and σ(EW) (Sect. 9.6) inherits that reliability.

9.2 Slit mode

The extracted fraction is the product of two couplings of Eq. (19): the slit width across dispersion and the finite extraction height along it. When the slit is *not* at the parallactic angle of Eq. (8), the differential refraction $\Delta R(\lambda)$ of Eq. (7) decentres the PSF in the slit per wavelength; SPETC applies the worst-case geometry (full offset perpendicular to the slit), reproducing the classical Filippenko [1982] blue-end losses (Fig. 2). A measured slit-width/resolving-power curve, when supplied, overrides the nominal R.

9.3 Slitless mode and the Star Analyser

In slitless spectroscopy the resolution element is set by the image quality, not by a slit: $\ell = \sqrt{(s/p)^2 + \ell_0^2 + \ell_{\rm AD}^2}$ combines seeing, the intrinsic grating LSF and the atmospheric-dispersion smear within one element (s = seeing FWHM, p = plate scale, ℓ_0 = intrinsic LSF in pixels, $\ell_{\rm AD}$ = dispersion smear in pixels). The sole extraction aperture is the cross-dispersion height.

The slitless sky is a full-bandpass background. The decisive physical difference from a slit is what happens to the *sky*. Dispersing a spatially *uniform* source leaves the detector illumination uniform: every wavelength of sky lands on every pixel, merely shifted, and the shifts of a uniform field are invisible. A slitless pixel therefore receives the sky integrated over the *entire* grating bandpass,

$$\dot{B}_{\rm pix} \;=\; p^2\, \eta\, A\, \varepsilon_g \int_{\lambda_{\rm min}}^{\lambda_{\rm max}} \mu_\lambda^{\rm sky}\, T_\lambda\, Q_\lambda\, O_\lambda\, \frac{\lambda}{hc}\, d\lambda, \tag{29}$$

with $\mu_\lambda^{\rm sky}$ the sky spectral surface brightness per arcsec^2, p^2 the pixel solid angle and ε_g the grating efficiency; the sky per element is $\dot{B}_{\rm pix}$ times the extraction pixels. This is the classical reason objective-grating spectra are sky-limited. Booking only one element's $\Delta\lambda$ of sky per pixel — as a slit would allow — underestimates the background by the ratio (bandpass width)/(element width), about two orders of magnitude for a Star Analyser covering 3300 Å with 28 Å elements; the v10.1 audit measured a $118\times$ deficit, corrected in v10.2 and regression-tested. Practical consequence: slitless S/N under a bright sky is far worse than a slit ETC intuition suggests, and the mode's advantage survives only for short exposures on brighter targets — which is precisely how Star Analysers are productively used.

For a transmission grating in the converging beam — the Star Analyser 100/200 geometry — the first-order deviation is $x \simeq L\, m\, n\, \lambda$, so the dispersion at the detector is

$$\mathcal{D} \;=\; \frac{10^4\, p_{\mu m}}{L_{\rm mm}\, n_{\rm mm}\, m} \quad [\text{Å/pixel}], \tag{30}$$

with n the groove density (100 or 200 mm^{-1}), L the grating-to-sensor distance and $m = 1$. An SA100 at $L = 42$ mm on 4.63 µm pixels gives $\mathcal{D} = 11.0$ Å/pixel, matching its published calibration; an SA200 at

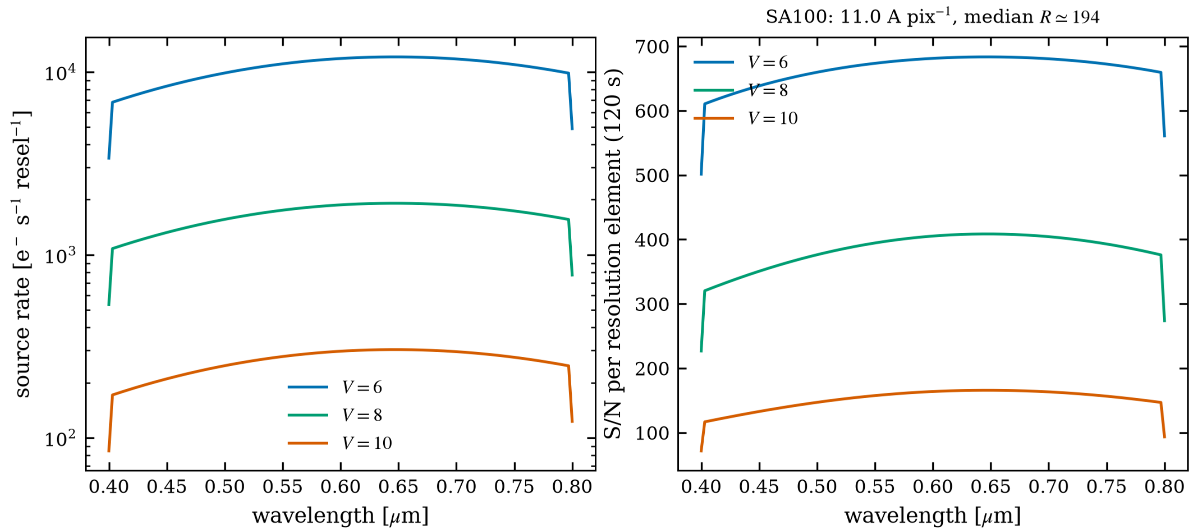

Figure 7: Star Analyser 100 predictions from the released code for a 200-mm f/5 telescope with a typical CMOS camera: detected source rate (left) and S/N per resolution element in 120 s (right) for $V = 6, 8, 10$ flat-spectrum targets.

the same distance gives half that. An optional scalar first-order grating efficiency multiplies source and sky alike. Under $2.5''$ seeing the effective resolving power is $R \simeq 100$–200 (Fig. 7), which is why the mode remains flagged *experimental* until validated against a calibrated instrument.

9.4 Detector footprint, diffraction orders, and optimal extraction

Given the detector dimensions $L \times W$ pixels (dispersion × cross-dispersion), the ETC reports whether the requested spectrum fits. The dispersion-direction length of a spectrum spanning $[\lambda_{\text{min}}, \lambda_{\text{max}}]$ is $(\lambda_{\text{max}} - \lambda_{\text{min}})/\mathcal{D}$ pixels in slitless mode (constant $\mathcal{D}$), and $s\,R\,\ln(\lambda_{\text{max}}/\lambda_{\text{min}})$ in slit mode, where s is the number of pixels per resolution element (since $\mathcal{D} = \lambda/(sR)$ there). For an objective grating the zero order (the undispersed star) sits at pixel 0 and order m places wavelength λ at $N_m(\lambda) = \lambda\,L_{\text{mm}}\,n\,m/(10^4 p_{\mu m})$ pixels from it; the ETC lists orders $m = 1, 2, 3$ and flags any whose footprint runs past L. This is exactly the overlap that a Star Analyser user manages by framing and, if needed, an order-blocking filter.

The Horne [1986] optimal-extraction algorithm weights each cross-dispersion pixel by P_x/σ_x^2 (profile over variance) rather than summing a fixed aperture, recovering the S/N of an infinite aperture. Its *noise-equivalent width* — the width of the top-hat that would give the same variance — is $1/\int P^2\,dx$, which for a Gaussian spatial profile of the seeing is $2\sqrt{\pi}\,\sigma = 1.51\,\text{FWHM}$. The ETC reports this as the optimal extraction width (arcsec and pixels); it is the extraction aperture that maximises S/N for the given seeing.

9.5 Slit-spectrograph geometry

Amateurs know their spectrograph by its hardware — groove density n, collimator and camera focal lengths $f_{\text{coll}}, f_{\text{cam}}$, slit width — not by R. In the Littrow approximation the grating equation gives $\sin\beta = mn\lambda/2$ and a reciprocal linear dispersion $d\lambda/dx = \cos\beta \cdot 10^7/(m\,n\,f_{\text{cam}})$ [Å/mm]. The resolution element on the detector is the narrower of the geometric slit image and the seeing image, both reimaged by $f_{\text{cam}}/f_{\text{coll}}$ (anamorphism 1 in Littrow):

$$\Delta\lambda = \min(w_{\text{slit}}, w_{\text{seeing}})\,\frac{f_{\text{cam}}}{f_{\text{coll}}}\,\frac{d\lambda}{dx}, \qquad R = \frac{\lambda}{\Delta\lambda}, \tag{31}$$

with the widths converted from arcsec at the telescope focal plane. The calculator reports whether the configuration is slit- or seeing-limited and feeds the ETC's resolving-power field, which remains editable;

an LHIRES III-like configuration (2400 lines/mm, $f = 200$ mm Littrow, 25-µm slit) returns $R \simeq 2 \times 10^4$ at Hα, in line with the instrument's specification.

Why the camera focal length does not change R. The three focal lengths of a spectrograph are distinct optical elements and must not be confused: the *telescope* focal length sets the plate scale at the slit (arcsec→mm), the *collimator* collimates the diverging beam onto the grating, and the *camera* refocuses the dispersed beam onto the detector. Substituting the reciprocal dispersion into $\Delta\lambda$ makes f_{cam} cancel exactly:

$$\Delta\lambda = \min(w_{\text{slit}}, w_{\text{seeing}}) \frac{f_{\text{cam}}}{f_{\text{coll}}} \frac{\cos\beta\, 10^7}{m\, n\, f_{\text{cam}}} = \min(w_{\text{slit}}, w_{\text{seeing}}) \frac{\cos\beta\, 10^7}{m\, n\, f_{\text{coll}}}, \tag{32}$$

so R depends on the slit (or seeing) width, the collimator focal length, the grating and the telescope focal length — but *not* on the camera focal length. This is a well-known and correct result, not a bug: the camera magnifies the slit image and the dispersion by the same factor, so their ratio (the resolving power) is invariant. What the camera focal length *does* set is the physical scale of the spectrum on the detector, hence the dispersion in Å/pixel and the number of pixels sampling one resolution element. Changing only f_{cam} in the helper therefore leaves R unchanged by design.

Two guards keep the entered R honest (v10.2). With the geometry supplied and the clamp option enabled, the engine itself limits the entered R to the geometry's value at the band centre — a physically impossible R can then no longer silently inflate every per-element prediction. Conversely, when the seeing disc is much *narrower* than the slit the delivered resolution is set by the star image, i.e. finer than the slit-limited R; the engine flags this case so the user knows the reported R is conservative.

9.6 Equivalent-width detectability

The spectroscopic S/N is turned into a science answer through the Cayrel [1988] relation for the equivalent-width uncertainty of a line of FWHM W sampled with pixels of width δx:

$$\sigma(\text{EW}) \simeq \frac{1.5\sqrt{W\,\delta x}}{(\text{S/N})_{\text{pixel}}}, \tag{33}$$

evaluated per wavelength with the resolution element as W and reported in mÅ in every result table. The practical criterion: a line is securely measurable when its expected equivalent width exceeds $\simeq 3\,\sigma(\text{EW})$ — e.g. $\sigma(\text{EW}) = 15$ mÅ means Hα variations of a Be star at the 50 mÅ level are detectable, while 20 mÅ photospheric lines are marginal.

9.7 Time series and reverse solving

Planning series evaluate the S/N reference wavelength bin at every time sample of the visibility window (full spectra only at the selected time and the plot-slider samples), with the sky model, airmass, refraction and parallactic angle recomputed per sample. Target-S/N requests use the closed-form solver of Sect. 8.1 per element and are checked against the saturation limit rather than being silently declared feasible.

10 Filter profiles and synthetic calibration

Professional passbands (Johnson–Cousins, Sloan, 2MASS, ...) ship as SVO Filter Profile Service VOTables [Rodrigo and Solano, 2020]. Amateur filters (Astrodon, Astronomik, Baader, Optolong, Chroma RGB and narrowband sets) are absent from SVO, but a measured or digitized transmission curve is all that is required: the zero points are *computed*, not measured. The bundled `make_filter_profile.py` applies the same synthetic convention SVO itself uses:

- **AB**: $Z_\nu = 3631$ Jy at every wavelength by definition [Oke and Gunn, 1983];

Table 4: Validation of the synthetic calibration against the shipped SVO 2MASS.H profile (whose declared values are independent of this code).

Quantity	SVO declared	Computed here	Difference
Pivot wavelength [Å]	16494.9	16494.7	-0.00%
Photon-weighted wavelength [Å]	16460.6	16423.8	-0.2%
Central wavelength [Å]	16506.1	16487.3	-0.1%
$F_\odot$ [erg cm^{-2} s^{-1} Å^{-1}]	22.44	22.33	-0.5%
Vega zero point [Jy]	1024.0	1030.4	$+0.6\%$

- **Vega**: the photon-weighted mean flux density of the CALSPEC Vega spectrum [Bohlin et al., 2014] through the filter,

$$Z_\nu = \frac{\int F_\lambda^{\text{Vega}} T_\lambda \, \lambda \, d\lambda}{\int \frac{c}{\lambda^2} T_\lambda \, \lambda \, d\lambda}, \tag{34}$$

  i.e. the flux a magnitude-zero star delivers in this band.

The tool writes full SVO-format VOTables with the complete photdm annotation and every characterising quantity measured from the curve by the SVO definitions (mean, pivot, effective and photon-weighted wavelengths, 1%-of-peak limits, effective width, FWHM, and the mean solar flux `Fsun` from the CALSPEC solar spectrum), so the produced files are drop-in interchangeable with SVO downloads and land on exactly the same photometric scale as the shipped professional profiles.

Table 4 validates the pipeline end-to-end on 2MASS.H: the sub-percent agreement (the residuals reflect SVO's slightly different Vega SED normalization) means a profile built from a digitized Astrodon or Astronomik curve is calibrated to the same standard as a professional filter — no laboratory measurement beyond the transmission curve itself is needed.

11 Expected performance and limits of validity

11.1 What to expect

- **Bright stars are scintillation-limited.** For the 358-mm reference telescope, a $V \simeq 5$ star in 60 s reaches S/N $\simeq 1.3 \times 10^3$, not the $\sim 2 \times 10^4$ a photon-only budget would claim (Fig. 6); the ceiling scales as $d^{2/3} X^{-1.75} \sqrt{t}$.

- **Faint limits are sky-limited.** Beyond $m \gtrsim \mu_{\text{sky}} - 2.5 \log_{10}(\pi r^2)$ the S/N falls as $10^{-0.4m}$ and doubling the exposure gains only $\sqrt{2}$.

- **Moonlight matters within $\sim 90°$.** Full Moon at 30° separation brightens the V sky by $\simeq$ 4–5 mag (Sect. 6), turning background-limited exposures from minutes into hours.

- **Slit spectroscopy away from the parallactic angle** loses tens of per cent at the blue end already at $X = 1.5$–2 (Fig. 2); the Results window reports q at every sample so the slit can be set correctly.

- **Star Analyser work** is photometrically generous (no slit losses) but limited to $R \sim 100$–200 by seeing; S/N per element of several hundred in minutes is realistic for $V \lesssim 8$ (Fig. 7).

11.2 Known approximations

The model is explicit about its accuracy floor: the U–K sky table and its colour interpolation are broad-band, and the twilight/daylight bridge is approximate; the extinction fallback table is a smooth broad-band baseline until a site-specific curve is loaded; scintillation uses the classical Young coefficient (site-specific coefficients from Osborn et al. 2015 can be substituted); atmospheric dispersion in slit mode assumes the worst-case orientation; extended sources are uniform discs; and slitless mode awaits validation against a calibrated instrument. Flat-field and fringing residuals, cosmic rays and sky-subtraction penalties are not modelled.

11.3 Numerical verification

Two test programs ship with the code: a synthetic smoke test that pins the scintillation-limited photometric S/N and the per-element spectroscopic S/N of the reference configuration, and a regression suite that verifies, among others, Eq. (15) against the published scattering function, the ecliptic symmetry of the zodiacal term, Eq. (5) against sec z, the Student-t Moffat coupling, the Filippenko dispersion scale, Eq. (30) for the SA100/SA200, and the extended-source area fractions.

12 A worked example, by hand

To connect the formulae to the numbers the program prints, this section computes one photometric case almost entirely by hand: a $V = 12.0$ (Vega) solar-type star, observed in V through a 200-mm f/5 Newtonian (secondary obstruction 70 mm, optics efficiency 0.85) with a typical CMOS camera (QE $\simeq$ 0.8 at 5500 Å, gain setting giving 1.0 e$^-$/ADU, read noise 1.5 e$^-$, pixel 3.76 µm), under 2.5″ seeing at airmass 1.2 from a suburban site (SQM = 20.0, elevation 300 m), for 60 s in a 5″-radius aperture.

1. Photon budget of the source. A $V = 0$ star delivers about 1.0×10^4 photons s^{-1} cm^{-2} Å^{-1}... more precisely, Eq. (1) with $Z_\nu = 3640$ Jy at 5500 Å gives $F_\lambda = 3.63 \times 10^{-9}$ erg s^{-1}cm^{-2}Å^{-1}, i.e. $F_\lambda/(hc/\lambda) \simeq 1005$ photons s^{-1}cm^{-2}Å^{-1}. At $V = 12$: $\times 10^{-0.4 \cdot 12} = 1.58 \times 10^{-5}$, so 1.59×10^{-2} photons s^{-1}cm^{-2}Å^{-1}.

2. Collecting area and effective bandwidth. $A = \pi(20.0^2 - 7.0^2)/4 = 275.7\,\mathrm{cm}^2$. The Bessell V band has an effective width of $\simeq 870$ Å. Through atmosphere ($T^X \simeq 0.87^{1.2} \simeq 0.85$ at V), optics (0.85) and QE (0.80):

$$\dot{S}_{\mathrm{tot}} \simeq 1.59 \times 10^{-2} \times 275.7 \times 870 \times 0.85 \times 0.85 \times 0.80 \;\simeq\; 2.2 \times 10^3 \ \mathrm{e^-\,s^{-1}}.$$

The program, integrating the real solar-type spectrum against the real V curve, reports $\simeq 2.1 \times 10^3$ e$^-$/s — the hand estimate is good to ten percent, which is the accuracy of using an effective width.

3. Aperture and sky. A 5″ radius is 2× the seeing FWHM: a Moffat ($\beta = 2.5$) profile puts 89% of the light inside (Sect. 7), so $\dot{S} = 0.89 \times 2100 \simeq 1.9 \times 10^3$ e$^-$/s, and $S = 1.1 \times 10^5$ e$^-$ in 60 s. The aperture covers $\pi\,25 = 78.5$ arcsec^2. SQM 20.0 means the sky is $m_{\mathrm{sky}} = 20.0$ mag/arcsec^2 in V, i.e. per arcsec^2 the sky is a 20.0-mag source: $\dot{B} = 2100 \times 10^{-0.4(20.0-12.0)} \times 78.5 \simeq 2.6 \times 10^2$ e$^-$/s, $B = 1.6 \times 10^4$ e$^-$ in 60 s. With a plate scale of 0.776″/pixel the aperture holds $N_{\mathrm{pix}} \simeq 130$ pixels.

4. Noise budget. Shot noise: $\sqrt{S+B} = \sqrt{1.1 \times 10^5 + 1.6 \times 10^4} = 355$ e$^-$. Read: $\sqrt{130} \times 1.5 = 17$ e$^-$. Quantization: $\sqrt{130} \times 1.0/\sqrt{12} = 3$ e$^-$. Scintillation, Eq. (10): $\sigma/I = 0.09 \times 20^{-2/3} \times 1.2^{1.75} \times e^{-300/8000}/\sqrt{120} = 1.5 \times 10^{-3}$, so $\sigma_{\mathrm{scint}} = 1.5 \times 10^{-3} \times 1.1 \times 10^5 = 165$ e$^-$. Total: $\sqrt{355^2 + 165^2 + 17^2 + 3^2} = 392$ e$^-$.

5. Result. $\mathrm{S/N} = 1.1 \times 10^5/392 \simeq 280$ — a differential precision of $1086/280 \simeq 3.9$ mmag per frame against an equal comparison star ($\times\sqrt{2}$: 5.5 mmag), fine for a 25-mmag exoplanet transit at two-minute cadence. Note where the noise lives: shot noise dominates, but scintillation is already 42% of it at $V = 12$ on a 20-cm aperture — brighten the star by three magnitudes and scintillation rules alone, which is why the S/N curve of Fig. 6 flattens. The peak pixel holds $\simeq 2100 \times 0.07 \times 60 \simeq 9 \times 10^3\,e^-$ (central-pixel fraction 0.07 for this sampling) plus sky and dark — far from a CMOS full well at low gain, so the 60-s frame is safe, as the program's saturation columns confirm.

Every number above is printed by one run of the program; the exercise shows there is no magic between the inputs and the outputs — only the chain of Sect. 2, executed carefully.

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